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4.4 Pythagoras' Theorem & Trigonometry

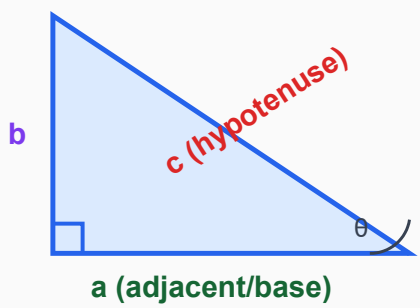
OCR GCSE Mathematics J560 — Component 4 (Geometry and Measures) — Papers 1, 2, 3

4.4.1 Pythagoras' Theorem

In any **right-angled triangle**, the square of the length of the hypotenuse (the side opposite the right angle) equals the sum of the squares of the other two sides.

$$a^2 + b^2 = c^2$$

where c is the hypotenuse.



Finding the hypotenuse:
$c = \sqrt{a^2 + b^2}$

Finding a shorter side:
$a = \sqrt{c^2 - b^2}$

Worked Example: A right-angled triangle has legs 5 cm and 12 cm. Find the hypotenuse.

$c = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = \mathbf{13}$ **cm**

Worked Example: The hypotenuse is 17 cm and one leg is 8 cm. Find the other leg.

$a = \sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = \mathbf{15}$ **cm**

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Common Pythagorean Triples

Triple	Multiples (×2, ×3, ...)
3, 4, 5	6,8,10 9,12,15 15,20,25
5, 12, 13	10,24,26
8, 15, 17	
7, 24, 25	

Pythagoras in 3D

For a cuboid with dimensions $l \times w \times h$, the space diagonal d is:

$$d = \sqrt{l^2 + w^2 + h^2}$$

Worked Example: A cuboid is 6 cm long, 4 cm wide, 3 cm tall. Find the length of the space diagonal.

First find the base diagonal: $\sqrt{6^2 + 4^2} = \sqrt{52}$
Then: $d = \sqrt{(\sqrt{52})^2 + 3^2} = \sqrt{52 + 9} = \sqrt{61} \approx \mathbf{7.81}$ **cm**

Exam Tip: Always identify the hypotenuse first — it is opposite the right angle and is the longest side. If the answer is not a whole number, leave it as a surd ($\sqrt{}$) unless asked to round.

4.4.2 SOHCAHTOA (Right-Angled Trigonometry)

For a right-angled triangle, the three basic trigonometric ratios relate an angle θ to the sides:

- **Opposite** — the side opposite to θ
- **Adjacent** — the side next to θ (not the hypotenuse)

- **Hypotenuse** — the longest side (opposite the right angle)

SOH

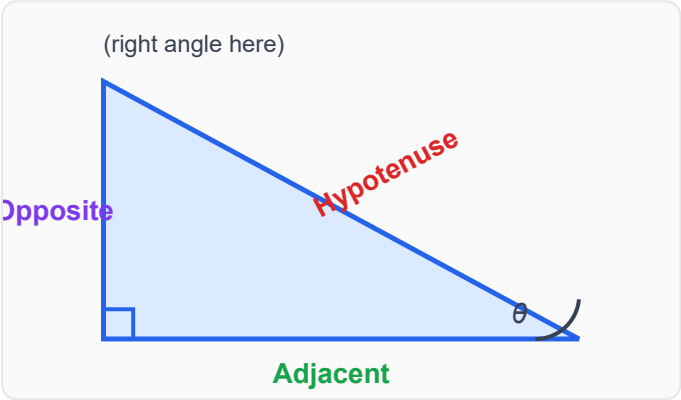
$$\sin\theta = \frac{\text{Opp}}{\text{Hyp}}$$

CAH

$$\cos\theta = \frac{\text{Adj}}{\text{Hyp}}$$

TOA

$$\tan\theta = \frac{\text{Opp}}{\text{Adj}}$$



Worked Example: Find a missing side: In triangle ABC, angle B = 35°, hypotenuse AB = 12 cm. Find BC (adjacent).

Using CAH: $\cos 35^\circ = \dfrac{BC}{12}$

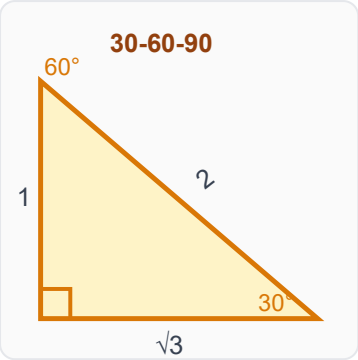
$BC = 12 \times \cos 35^\circ = 12 \times 0.8192 \approx \mathbf{9.83}$ **cm**

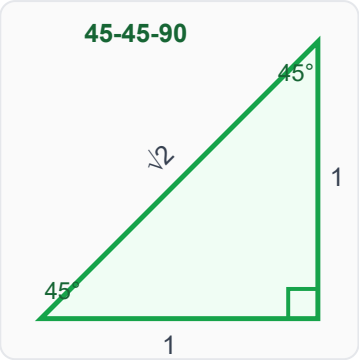
Worked Example: Find a missing angle: Opposite = 7 cm, hypotenuse = 10 cm. Find θ.

Using SOH: $\sin\theta = \dfrac{7}{10} = 0.7$

$\theta = \sin^{-1}(0.7) \approx \mathbf{44.4}^\circ$

Special Triangles

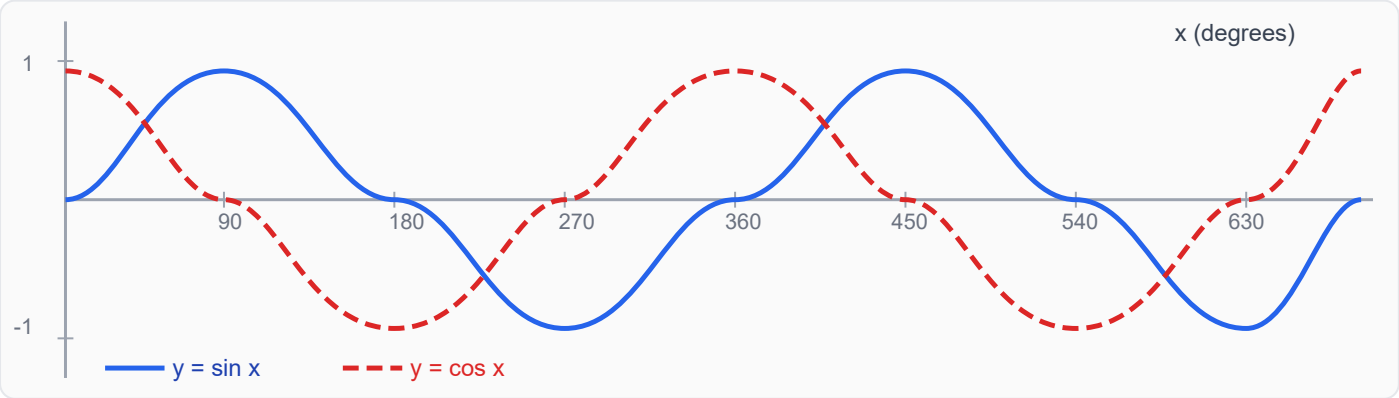




Angle	sin	cos	tan
0°	0	1	0
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	undefined

Exam Tip: When asked to give an exact value, use the special triangle values above — do not use a decimal approximation. On non-calculator papers, you will always get a special angle.

4.4.3 Trigonometric Graphs H



Graph	Period	Amplitude	Key features
$y = \sin x$	360°	1	Starts at (0,0); max at 90°; zero at 180°, 360°; min at 270°
$y = \cos x$	360°	1	Starts at (0,1); zero at 90°, 270°; min at 180°; max at 360°
$y = \tan x$	180°	Unbounded	Asymptotes at 90°, 270°; passes through (0,0), (45°,1), (135°, −1)

Symmetry and Solutions

- $\sin(180^\circ - x) = \sin x$ — sine is symmetric about 90°
- $\cos(360^\circ - x) = \cos x$ — cosine is symmetric about 0°/360°
- $\cos(-x) = \cos x$ — cosine is an even function
- If $\sin\theta = k$, the second solution in $[0^\circ, 360^\circ]$ is $180^\circ - \theta$
- If $\cos\theta = k$, the second solution in $[0^\circ, 360^\circ]$ is $360^\circ - \theta$

Worked Example: Solve $\sin\theta = 0.6$ for $0^\circ \leq \theta \leq 360^\circ$.

$\theta = \sin^{-1}(0.6) = 36.87^\circ$
Second solution: $180^\circ - 36.87^\circ = 143.13^\circ$
$\theta \approx 36.9^\circ$ **or** 143.1°

4.4.4 Sine Rule H

For any triangle with sides a, b, c opposite to angles A, B, C respectively:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Rearranged for finding angles:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Use the Sine Rule when you know: two angles and one side (AAS / ASA), or two sides and a non-included angle (SSA).

Worked Example: In triangle ABC: $a = 8$ cm, $A = 40^\circ$, $B = 70^\circ$. Find b.

$$\frac{b}{\sin 70^\circ} = \frac{8}{\sin 40^\circ}$$

$$b = \frac{8 \times \sin 70^\circ}{\sin 40^\circ} = \frac{8 \times 0.9397}{0.6428} \approx \mathbf{11.7\ cm}$$

Worked Example: Find angle B given $b = 12$ cm, $a = 9$ cm, $A = 55^\circ$.

$$\frac{\sin B}{12} = \frac{\sin 55^\circ}{9}$$

$$\sin B = \frac{12 \times \sin 55^\circ}{9} = \frac{12 \times 0.8192}{9} = 1.092...$$

Since $\sin B > 1$, no solution exists — the triangle is impossible with these measurements.

The Ambiguous Case (SSA)

When given two sides and a non-included angle, there may be **0, 1, or 2 valid triangles**. If $\sin B < 1$, check whether B (acute) and $180^\circ - B$ (obtuse) both give a valid triangle (i.e. sum of angles < 180°).

Exam Tip: For angles, use the form $\frac{\sin A}{a}$ (sine over side). For sides, use $\frac{a}{\sin A}$ (side over sine). Only use two parts of the rule at once — pick the pair with the most knowns.

4.4.5 Cosine Rule H

For any triangle:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Rearranged to find an angle:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Use the Cosine Rule when you know: two sides and the included angle (SAS), or all three sides (SSS).

Worked Example: Sides $b = 7$, $c = 10$, included angle $A = 50^\circ$. Find a.

$a^2 = 7^2 + 10^2 - 2(7)(10)\cos 50^\circ$
$a^2 = 49 + 100 - 140 \times 0.6428 = 149 - 89.99 = 59.01$
$a = \sqrt{59.01} \approx \mathbf{7.68}$ **cm**

Worked Example: Three sides $a = 5$, $b = 7$, $c = 9$. Find angle A.

$$\cos A = \frac{7^2 + 9^2 - 5^2}{2 \times 7 \times 9} = \frac{49 + 81 - 25}{126} = \frac{105}{126} = 0.8333$$

$A = \cos^{-1}(0.8333) \approx \mathbf{33.6}^\circ$

Exam Tip: Never use the Cosine Rule for a right-angled triangle — Pythagoras or SOHCAHTOA is simpler. The Cosine Rule is for non-right-angled triangles only.

4.4.6 Area of a Triangle H

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For any triangle with two sides a and b and included angle C:

$$A = \frac{1}{2}ab\sin C$$

Worked Example: Triangle with sides 8 cm and 11 cm, included angle 63°. Find the area.

$A = \frac{1}{2} \times 8 \times 11 \times \sin 63^\circ = 44 \times 0.891 \approx \mathbf{39.2}$ **cm²**

Worked Example: The area of a triangle is 30 cm². Two sides are 10 cm and 9 cm. Find the included angle.

$30 = \frac{1}{2} \times 10 \times 9 \times \sin C$
$\sin C = \dfrac{30}{45} = 0.6\overline{6}$
$C = \sin^{-1}(0.6\overline{6}) \approx \mathbf{41.8}^\circ$

Choosing the Right Rule

Given	Find	Rule to use
Right angle + 2 sides	Side or angle	Pythagoras / SOHCAHTOA
AAS or ASA (2 angles, 1 side)	Side or angle	Sine Rule
SSA (2 sides, non-included angle)	Angle or side	Sine Rule
SAS (2 sides, included angle)	Third side	Cosine Rule
SSS (3 sides)	Any angle	Cosine Rule
SAS (2 sides, included angle)	Area	$\frac{1}{2}ab\sin C$

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